TASK SPECIFICATION: P7-06 PHASE P7 — NOTIFICATIONS, GOVERNANCE & TELEMETRY ANALYTICS

Campus Analytics & Reporting Telemetry Engine

Git Branch: `feature/P7-06-analytics` · Target Pull Request: feature/P7-06-analytics

Core Objective & Technical Scope:
Real-time analytics queries calculating hourly spot utilization, peak arrival curves, tutoring volume, and carbon emission offsets.

Priority Level	Low	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P7-06-analytics`
Upstream Dependencies	P5-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author optimized SQL analytical aggregation queries.
2. Create AnalyticsController exposing metrics endpoints.
3. Build interactive chart visualizations in frontend (peak hours, occupancy curves).
4. Verify performance under large historical datasets.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `analytics/**`
- `frontend/src/features/analytics/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- All analytics metrics derived from actual database records with zero synthetic mock data.
- Aggregation queries execute in under 100ms.

VERIFICATION & TEST COVERAGE STRATEGY

- AnalyticsRepositoryTest
- AnalyticsControllerTest
- ChartComponentTest